A Pre-Massage Multimodal Physiological State Assessment and Expected Fatigue Relief Prediction Framework

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ABSTRACT

An individual's physiological condition before massage may affect both the fatigue-related response during massage and the physiological changes observed after massage. However, methods that rely on massage-stage electroencephalography or post-massage physiological measurements cannot support personalized decision-making before massage begins. To address this limitation, this paper proposes a pre-massage multimodal prediction framework with two complementary outputs.

The first output is the Baseline Physiological State Index (BPSI), which summarizes an individual's current physiological condition before massage using anthropometric information, electrocardiographic measurements, blood pressure, and related physiological indicators. The second output is the Expected Fatigue Relief Index (EFRI), which quantifies the expected fatigue-related improvement after massage. EFRI is constructed during the training stage using fatigue-related EEG responses during massage and physiological changes after massage, whereas only pre-massage information is used during prediction. The proposed framework consists of a body-information branch, a pre-massage physiological-information branch, a shared feature fusion module, and two output heads for BPSI and EFRI. Several BPSI weighting strategies, including equal weighting, principal component analysis, partial least squares, regularized regression, and data-driven neural weighting, are compared. In addition, regularized regression, ensemble learning, multilayer perceptrons, and a dual-output neural network are evaluated under subject-level cross-validation and strict information leakage control. The objective of this study is to investigate whether pre-massage multimodal physiological information can support current-state assessment and expected fatigue-related response prediction before massage begins.

INTRODUCTION

Fatigue is a multidimensional physiological and cognitive state that may be reflected in changes in cardiovascular activity, blood pressure, autonomic regulation, attention, and brain activity. In practical health-care and wellness applications, accurate fatigue assessment is important for selecting appropriate interventions and evaluating individual responses to treatment.

Massage is widely used as a non-pharmacological intervention for relaxation, stress reduction, and fatigue relief. However, individuals may exhibit different physiological responses to the same massage procedure. The pre-massage physiological condition may affect the degree of relaxation, the change in fatigue-related EEG activity, and the recovery of cardiovascular or circulatory indicators after massage. Therefore, an assessment method that can estimate the individual's initial state and potential benefit before massage may support more personalized intervention decisions.

Existing fatigue-assessment studies have mainly focused on either physiological signals acquired during fatigue-inducing tasks or EEG signals recorded during cognitive or relaxation conditions. In massage-related studies, post-massage physiological changes are commonly used to evaluate intervention effects. Although these approaches can provide useful information, they generally require data collected during or after the intervention. Consequently, they cannot directly support decision-making before massage begins.

To address this limitation, this study develops a pre-massage multimodal prediction framework with two outputs. The first output, the Baseline Physiological State Index (BPSI), describes the individual's integrated physiological condition immediately before massage. The second output, the Expected Fatigue Relief Index (EFRI), estimates the individual's potential fatigue-related improvement after massage. During model development, EFRI is constructed using massage-stage EEG responses and post-massage physiological changes. During actual prediction, however, only pre-massage information is used.

The main contributions of this study are as follows:

- A Baseline Physiological State Index is constructed to summarize the individual's pre-massage physical and physiological condition.
- An Expected Fatigue Relief Index is developed from fatigue-related EEG responses and post-massage physiological changes.
- A pre-massage dual-output prediction framework is proposed to generate both BPSI and EFRI before massage begins.
- Multiple weighting strategies and machine-learning and deep-learning models are systematically compared.
- Subject-level validation, information leakage control, sensitivity analysis, and interpretability analysis are employed to evaluate the reliability and practical value of the proposed framework.

The remainder of this paper is organized as follows. Section II reviews previous studies on physiological and EEG-based fatigue assessment, massage-related physiological response assessment, and data-driven prediction. Section III introduces the proposed pre-massage multimodal prediction framework. Section IV presents the experimental design and results. Finally, Section V concludes this paper.

1 RELATED WORKS

1.1 Physiological and EEG-Based Fatigue Assessment

Physiological fatigue assessment commonly uses heart rate, heart rate variability, blood pressure, blood oxygen saturation, microcirculation, and other cardiovascular indicators. These measurements provide information related to autonomic regulation and physiological load. However, the interpretation of absolute physiological values may be affected by individual differences in age, body composition, baseline cardiovascular condition, and health status.

EEG-based methods provide complementary information about central nervous system activity, attention, mental fatigue, and mind wandering. EEG-derived fatigue values, attention scores, mind-wandering frequency, and related indicators have been used to describe changes in cognitive and mental states. Compared with peripheral physiological measurements, EEG may provide more direct information about changes in brain activity. However, EEG acquisition usually requires additional equipment and may not be convenient for routine pre-intervention prediction.

Previous studies have also demonstrated that fatigue is a multidimensional state and cannot be fully represented by a single physiological variable. Therefore, the integration of multiple physiological and cognitive indicators may provide a more comprehensive description of an individual's condition.

1.2 Massage-Related Physiological Response Assessment

Massage-related studies generally evaluate intervention effects using changes in physiological measurements before and after massage. Common variables include heart rate, systolic blood pressure, diastolic blood pressure, microcirculation, oxygen saturation, and ECG-derived fatigue indicators. EEG measurements collected during massage can also be used to characterize relaxation, attention, stress reduction, and fatigue-related responses.

However, the magnitude of a post-massage physiological change does not necessarily indicate a stronger fatigue improvement. The direction and physiological meaning of each variable must be considered. For example, a decrease in a fatigue-related index may indicate improvement, whereas an increase in heart rate or blood pressure may have different meanings depending on the individual's baseline state and measurement context.

Most existing studies focus on describing physiological responses after massage. Fewer studies attempt to predict these responses using only information available before massage. This limitation makes it difficult to provide personalized decisions before the intervention begins. The proposed framework addresses this problem by using massage-stage EEG and post-massage physiological measurements only for target construction during model development, while using pre-massage information as the only prediction input.

PROPOSED NETWORK

The objective of the proposed framework is to generate two outputs before massage begins: the current Baseline Physiological State Index and the Expected Fatigue Relief Index.

1.3 Preprocessing and Feature Organization

The available data are organized into three groups:

1. Pre-massage body-related information;
2. Pre-massage ECG, blood pressure, and related physiological information;
3. Massage-stage EEG and post-massage physiological measurements used for target construction.

The pre-massage body-related variables include age, sex, height, weight, body fat percentage, body mass index, basal metabolic rate, visceral fat, muscle mass, body water percentage, body score, and body age.

The pre-massage physiological variables include heart rate, microcirculation, ECG-derived fatigue index, blood oxygen saturation, health score, systolic blood pressure, and diastolic blood pressure.

The massage-stage EEG variables include EEG fatigue value, mean attention, maximum attention, meditation duration, meditation proportion, mind-wandering frequency, stress-relief index, comprehensive score, and mean mindfulness index.

The post-massage physiological variables include the same physiological measurements collected after massage. For each physiological variable, the post-massage response is calculated relative to the corresponding pre-massage measurement.

Numerical features are standardized using the mean and standard deviation calculated from the training data. Categorical variables are encoded using *[binary or one-hot encoding]*. Missing values are handled using *[the selected method]*, and abnormal values are detected using *[the selected rule]*.

To prevent information leakage, all preprocessing parameters are estimated independently within each training fold. The parameters estimated from the training fold are then applied to the validation and test folds.

Let the pre-massage input vector be defined as

$$\mathbf{x} = \left[\mathbf{x}_{\text{body}}, \mathbf{x}_{\text{phys}}\right], \tag{1}$$

where $\mathbf{x}_{\text{body}}$ represents body-related variables and $\mathbf{x}_{\text{phys}}$ represents pre-massage physiological variables.

Baseline Physiological State Index(BPSI)

The Baseline Physiological State Index is designed to describe the individual's integrated physiological condition before massage. BPSI is not treated as an independent gold-standard measure of fatigue. Instead, it is defined as a composite index based on multiple pre-massage variables.

Let $\tilde{x}_j$ denote the standardized and direction-aligned value of the j-th pre-massage feature. BPSI can be expressed as

$$\text{BPSI} = \sum_{j=1}^{p} w_j \tilde{x}_j, \tag{2}$$

where w_j is the data-driven weight of the j-th feature and p is the number of input features.

The primary BPSI should use a state-oriented weighting strategy. Principal component analysis or factor analysis can be used to derive weights from the statistical structure of the pre-massage features. The resulting BPSI can be linearly rescaled to a range of 0–100 for presentation.

The following weighting strategies are compared:

- Equal weighting;
- Principal-component-based weighting;
- Factor-analysis-based weighting;
- Partial-least-squares weighting;
- Elastic-Net-based weighting;
- Data-driven neural weighting.

PCA- and factor-analysis-based BPSI mainly describe the statistical structure of the current physiological state. PLS- and Elastic-Net-based BPSI are response-oriented alternatives because they use EFRI-related information during model construction. Therefore, their interpretation should be distinguished from the primary state-oriented BPSI.

A higher BPSI should indicate a higher physiological load only after the direction of every input variable has been aligned. Variables for which a higher value indicates better physiological status should be reverse-coded before aggregation.

Expected Fatigue Relief Index(EFRI)

The Expected Fatigue Relief Index represents the fatigue-related improvement expected after massage. EFRI is constructed during the model-development stage from massage-stage EEG responses and post-massage physiological changes.

The EEG-related component may include EEG fatigue value, mean attention, and mind-wandering frequency. Meditation duration, meditation proportion, mindfulness index, stress-relief index, and comprehensive score should not automatically be regarded as objective fatigue labels. Their inclusion should be supported by statistical analysis or sensitivity analysis.

The post-massage physiological response variables include changes in heart rate, systolic blood pressure, diastolic blood pressure, microcirculation, ECG-derived fatigue index, blood oxygen saturation, and health score.

Let $\mathbf{e}$ denote the aligned EEG response variables and let $\mathbf{r}$ denote the aligned post-massage physiological responses. EFRI is generally defined as

$$\mathrm{EFRI} = \sum_{k=1}^{q} v_k e_k + \sum_{l=1}^{m} u_l r_l, \tag{3}$$

where v_k and u_l are data-driven weights.

The direction of each variable must be aligned before aggregation. For example, a reduction in fatigue value or mind-wandering frequency may represent improvement, whereas an increase in an improvement-related score may represent improvement.

Three EFRI definitions are evaluated:

1. EEG-only EFRI;
2. Post-massage physiological-response EFRI;
3. Joint EEG–physiological EFRI.

EFRI is regarded as a composite fatigue-related response index rather than an absolute fatigue ground truth. If no control or sham-massage group is available, EFRI should not be interpreted as a strict causal treatment effect.

1.4 Dual-Output Prediction Network

The proposed model consists of two input branches. The first branch processes body-related information, and the second branch processes pre-massage physiological information.

The two branches extract modality-specific features:

$$\mathbf{h}_{\text{body}} = f_{\text{body}}(\mathbf{x}_{\text{body}}), \tag{4}$$

$$\mathbf{h}_{\text{phys}} = f_{\text{phys}}(\mathbf{x}_{\text{phys}}). \tag{5}$$

The extracted features are concatenated and processed by a shared fusion layer:

$$\mathbf{h} = f_{\text{fusion}}\left(\mathbf{h}_{\text{body}}, \mathbf{h}_{\text{phys}}\right). \tag{6}$$

Two output heads are connected to the shared feature representation:

$$\widehat{\text{BPSI}} = g_{\text{BPSI}}(\mathbf{h}), \tag{7}$$

$$\widehat{\text{EFRI}} = g_{\text{EFRI}}(\mathbf{h}). \tag{8}$$

The joint training objective is

$$\mathscr{L} = \lambda_{\text{B}}\mathscr{L}_{\text{BPSI}} + \lambda_{\text{E}}\mathscr{L}_{\text{EFRI}} + \lambda_{\text{R}}\mathscr{L}_{\text{reg}}, \tag{9}$$

where $\mathscr{L}_{\text{BPSI}}$ and $\mathscr{L}_{\text{EFRI}}$ denote the losses of the two output heads, $\mathscr{L}_{\text{reg}}$ denotes the regularization term, and λ_{B}, λ_{E}, and λ_{R} are the corresponding coefficients.

If BPSI is directly calculated from a learned weighting rule, the BPSI head can be regarded as an auxiliary output. In practical deployment, BPSI may be calculated directly from the pre-massage input variables, while EFRI is predicted by the EFRI output head.

The model does not use massage-stage EEG or post-massage physiological measurements as input during prediction.

1.5 Model Training and Validation

The following models are compared:

1. Ridge regression;
2. Elastic Net;
3. Random forest;
4. XGBoost;
5. Single-output multilayer perceptron;
6. Proposed dual-output multimodal multilayer perceptron.

The dual-output network is designed as a lightweight network because the expected sample size is approximately 500 or more. Excessively deep architectures are avoided to reduce overfitting. Dropout, weight decay, and early stopping are used when necessary.

All data are divided at the participant level. Standardization, feature selection, BPSI weighting, EFRI weighting, and hyperparameter tuning are performed only within the training data.

Nested cross-validation is used for model selection and unbiased performance estimation. The final model is retrained on the complete training set using the selected hyperparameters and evaluated on a held-out test set.

Table 1. Participant characteristics and main pre-massage variables.

Variable	Mean ± SD	Range
Age	[value]	[value]
BMI	[value]	[value]
Heart rate	[value]	[value]
SBP	[value]	[value]
DBP	[value]	[value]
ECG fatigue index	[value]	[value]

2 EXPERIMENTAL RESULTS

2.1 Dataset and Data Acquisition

The final dataset contains *[N]* complete participant records. The dataset includes *[n]* males and *[n]* females, with an average age of *[mean]* years and an age range of *[range]* years.

The data acquisition procedure consists of three stages:

1. Pre-massage measurement of body-related and physiological variables;
2. EEG measurement during massage;
3. Post-massage measurement of physiological variables.

Table 1 summarizes the participant characteristics and the main physiological variables.

2.2 Performance Metrics

The performance of EFRI prediction is evaluated using mean absolute error, root mean squared error, coefficient of determination, and correlation coefficient.

The mean absolute error is defined as

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N} |y_i - \hat{y}_i| \,. \tag{10}$$

The root mean squared error is defined as

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}. \tag{11}$$

The coefficient of determination is defined as

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}. \tag{12}$$

For BPSI, evaluation focuses on weighting stability, feature-selection frequency, direction consistency, physiological interpretability, and association with EFRI. Since BPSI is a constructed index rather than an independent gold-standard label, its performance should not be evaluated only using prediction accuracy.

2.3 Verification of BPSI Weighting Strategies

This subsection evaluates whether the selected weighting strategy can provide a stable and interpretable BPSI.

The following weighting strategies are compared:

- Equal weighting;
- PCA-based weighting;

Table 2. Comparison of different BPSI weighting strategies.

Strategy	Stability	EFRI association	Interpretability	Features
Equal weighting	[value]	[value]	[value]	[value]
PCA	[value]	[value]	[value]	[value]
Factor analysis	[value]	[value]	[value]	[value]
PLS	[value]	[value]	[value]	[value]
Elastic Net	[value]	[value]	[value]	[value]
Neural weighting	[value]	[value]	[value]	[value]

- Factor-analysis-based weighting;
- PLS-based weighting;
- Elastic-Net-based weighting;
- Neural-network-based weighting.

The comparison focuses on the following aspects:

1. Stability of weights across cross-validation folds;
2. Consistency of the signs of the weights;
3. Frequency with which variables are selected;
4. Correlation between BPSI and EFRI;
5. Physiological interpretability of the resulting index.

2.4 Verification of the Dual-Output Prediction Framework

This subsection verifies the effectiveness of the proposed dual-output design.

The following model structures are compared:

1. Independent EFRI prediction model without BPSI supervision;
2. Shared feature layer with independent BPSI and EFRI output heads;
3. Dual-output model without body-information branch;
4. Dual-output model without physiological-information branch;
5. Complete dual-branch dual-output model.

The purpose of this experiment is to determine whether the joint learning of BPSI and EFRI improves EFRI prediction while preserving the conceptual distinction between current physiological state and expected fatigue-related response.

The use of EFRI as a direct gating signal for BPSI is not adopted as the primary design because it may cause information overlap between the two outputs. A response-guided BPSI can be included as an additional comparison strategy, but it should be interpreted as a response-oriented index rather than a pure current-state index.

2.5 Ablation Experiments

Ablation experiments are conducted by removing different input groups or model components.

The following input configurations are compared:

1. Body-related variables only;
2. Pre-massage physiological variables only;
3. Body-related and physiological variables combined;
4. Combined inputs without ECG-derived fatigue index;
5. Combined inputs without blood pressure variables;
6. Combined inputs without body-composition variables.

The following model-component ablations are also considered:

1. Removing the body-information branch;
2. Removing the physiological-information branch;
3. Removing the shared fusion layer;
4. Removing the BPSI auxiliary output;
5. Removing regularization.

The ablation results are used to determine the contribution of each modality and model component to EFRI prediction.

2.6 Effects of Training Parameters

The effects of training parameters are evaluated while keeping the model structure unchanged.

The following parameters are analyzed:

1. Hidden-layer dimension;
2. Learning rate;
3. Batch size;
4. Dropout rate;
5. Weight-decay coefficient;
6. BPSI loss coefficient λ_B;
7. EFRI loss coefficient λ_E;
8. Number of training epochs.

The parameter values should be selected according to the validation performance rather than the test performance. The final parameter configuration is determined using only the training and validation data.

2.7 Comparison With Existing Machine-Learning and Deep-Learning Models

The proposed dual-output model is compared with regularized regression, ensemble-learning models, and single-output neural networks.

The comparison models include Ridge regression, Elastic Net, random forest, XGBoost, and a single-output multilayer perceptron. The proposed model is evaluated using the same input features, data split, preprocessing procedure, and evaluation metrics.

The main comparison should focus on EFRI prediction because EFRI is the deployment-oriented prediction target. BPSI is evaluated through its construction stability, interpretability, and physiological relevance.

Table 3. Performance comparison of different prediction models.

Model	MAE	RMSE	R^2	Correlation
Ridge	[value]	[value]	[value]	[value]
Elastic Net	[value]	[value]	[value]	[value]
Random Forest	[value]	[value]	[value]	[value]
XGBoost	[value]	[value]	[value]	[value]
MLP	[value]	[value]	[value]	[value]
Proposed model	[value]	[value]	[value]	[value]

2.8 Statistical Significance

To determine whether the performance differences between models are statistically meaningful, fold-level prediction errors are compared using paired statistical tests.

The selected statistical test is *[paired t-test, Wilcoxon signed-rank test, or another appropriate test]*. The significance level is set to *[e.g., 0.05]*. When multiple models are compared, multiple-comparison correction is applied using *[the selected correction method]*.

Bootstrap confidence intervals are also calculated for MAE, RMSE, R^2, and the correlation coefficient.

2.9 Generalization and Robustness Tests

The robustness of the proposed framework is evaluated using repeated cross-validation, bootstrap resampling, alternative train-test splits, and different EFRI construction strategies.

Three EFRI definitions are compared:

1. EEG-only EFRI;
2. Post-massage physiological-response EFRI;
3. Joint EEG–physiological EFRI.

If the dataset size permits, subgroup analyses are conducted according to age, sex, and baseline BPSI level.

The robustness analysis is used to determine whether the model performance depends on a particular data split, subgroup, or target-construction method.

2.10 Computational Complexity

The computational complexity of the proposed model is evaluated using the number of trainable parameters, training time, inference time, and memory requirements.

Because the proposed framework is designed for pre-massage decision-making, inference time is more important than the computational cost of target construction. EEG and post-massage physiological measurements are required during model development but are not required during deployment.

The following indicators are reported:

- Number of trainable parameters;
- Training time;
- Inference time per participant;
- Memory usage;
- Model size.

The computational results are compared with those of the other prediction models.

CONCLUSIONS

This paper proposes a pre-massage multimodal physiological prediction framework with two complementary outputs. The Baseline Physiological State Index describes the individual's current physiological condition before massage, whereas the Expected Fatigue Relief Index estimates the expected fatigue-related improvement after massage.

The proposed framework uses body-related information and pre-massage physiological measurements as the only prediction inputs. Massage-stage EEG and post-massage physiological measurements are used only during model development to construct the EFRI target. Therefore, the framework can support decision-making before massage begins.

Several data-driven weighting strategies, machine-learning models, and deep-learning models are compared. The dual-output network provides a unified architecture for current-state assessment and response prediction while preserving the distinction between BPSI and EFRI.

The proposed method is expected to provide a foundation for personalized massage decision-making. Future work should include external validation, larger and more diverse populations, different massage protocols, and controlled studies with sham or non-massage groups.

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REFERENCES